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Optimizing methodologies of hybrid renewable energy systems powered reverse osmosis plants

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ABSTRACT

Based on a systematic and bibliometric review, the combined use of Portfolio Theory (PT) and Particle Swarm Optimization (PSO) is proposed for the energy management of a Hybrid Renewable Energy System (HRES) powered reverse osmosis (RO) plant. PT-PSO combined use aims a Multi-Objective Optimization (MOO), decreasing the Drinking Water Cost (DWC) and increasing the system reliability and components lifetime. HRES optimization is crucial to meet the RO demand, considering that the sources intermittence can lead to a reduction of the lifetime of RO membranes and other components. Initially, a systematic and bibliometric review is conduced to identify the strengths and weakness of the existing HRES optimization methodologies. The review reveals that HOMER is the most used software for hybrid plants sizing and optimization; however, HOMER shows some constraints, such as a limited range of the components nominal power and a high computational cost associated with the inclusion of hydrogen production and storage. In contrast, optimization methods based on PSO show a lower computational cost, delivering robust results. Additionally, PSO can be used with other methodologies, such as the PT; traditionally applied in the economy sector to improve investment portfolios, PT recently has been used in the energy sector for wind and solar forecasting; PT use for HRES energy management is innovative.

1. Introduction

World's population is expected to reach 9.7 billion inhabitants by 2050 [1]; such population growth, combined with an increasing quality of life in most of the countries, makes electricity generation a central concern [2]. By 2050 the world electricity generation is expected to have an increase of 123%–150% [3]. To face such context, the world electricity sector, responsible for a large part of the greenhouse gases (GHG) emissions, is changing to use low carbon sources, mainly solar and wind, which are gaining an increasing participation [4,5]. Motivated by the need of a sustainable energy supply, fully renewable based power plants are increasingly being proposed in hybrid configurations [6]. Due to the solar and wind intermittence, a resources combination is a strategy for a greater generation reliability [7]. Among the various applications, an important use of such hybrid systems is to power reverse osmosis (RO) plants [8,9].

Currently, drinking water is not accessible to approximately 29% of the world's population [10]. According to the World Health Organization, half of the world's population will be in water-stressed areas by 2025 [11]. As the need of potable water supply increases, a fast development is observed in the desalination sector during the last years [12,13]; the desalination process is being improved, considering control strategies and energy costs [14]. One of the main desalination methods is RO [15]; with proper location, design and operation, RO plants can help to decrease energy requirements and environmental effects [16].

In the natural osmotic process, a solvent flows spontaneously through a semipermeable membrane, which separates two solutions with different concentrations of solute. The solvent flows spontaneously from the solution with low concentration to the one with high concentration of solute, until the system enters osmotic equilibrium, where it remains, considering the process irreversibility, as long as there is no energy input into the system [17]. The reverse process occurs when a pressure greater than the osmotic pressure is applied; hence, for sea and brackish water desalination, the solvent (water) flows from the solution with high concentration to the one with low concentration of solute (salt). Additionally, the membrane blocks the passage of about 99% of

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Nomenclature

AI	Artificial Intelligence
ANN	Artificial Neural Network
AS	Ant System
BAO	Bat Algorithm Optimization
BSO	Bee Swarm Optimization
BSS	Battery Storage System
BTU	Brine Treatment Unit
BWRO	RO plant for brackish water desalination
CA	Compressed Air
CG	Coal Gasification
CHP	Combined heat and power
COE	Cost of Electricity
CS	Cuckoo Search
CSP	Concentrated Solar Power
DA	Division Algorithm
DG	Diesel Generator
DWC	Drinking Water Cost
ED	Electrodialysis
EMP	Extended Mathematical Programming
FD	Freeze Desalination
FO	Fuel Oil
GA	Genetic Algorithm
GHG	Greenhouse Gases
GTE	Geothermal
GWO	Grey Wolf Optimization
HAS	Hybrid Self-Adaptive
HCS	Hybrid Chaotic Search
HGAPSO	Hybrid Genetic Algorithm Particle Swarm Optimization
HOMER	Hybrid Optimization Model for Multiple Energy Resources
HRES	Hybrid Renewable Energy System
HSAO	Harmony Search Algorithm Optimization
HSCSA	Improved Harmony Search-based Chaotic Simulated Annealing
LCC	Life Cycle Cost
LCOE	Levelized Cost of Electricity
LCOW	Levelized cost of water
LD	Load
LPSP	Loss of Power Supply Probability
MCDM	Multi Criteria Decision Making
MED	Multi Effect Distillation
MES	Multi-Energy Systems
MILP	Mixed Integer Linear Programming
MINLP	Mixed Integer Nonlinear Program
MO	Multi Objective
MOMVO	Multi-Objective Multi-Verse Optimization
MOO	Multi-Objective Optimization
MPSO	Modified Particle Swarm Optimization
MSF	Multi-Stage Flash
NG	Natural Gas
NPC	Net Present Cost
NSGA-II	Non-Dominated Sorting Genetic Algorithm-II
ORC	Organic Rankine Cycle
PG	Power generation
PT	Portfolio Theory
PLPSP	Potential loss of power supply probability
PSO	Particle Swarm Optimization
PTC	Parabolic trough collector
PV	Photovoltaic
RAVEN	Risk Analysis Virtual Environment
RF	Renewable Factor
RO	Reverse Osmosis
ROSA	Reverse Osmosis System Analysis
SAA	Simulated Annealing Algorithm
SAO	Search Algorithm Optimization
SEC	Specific Energy Consumption
SSA	Salp Swarm Algorithm
SSO	Social Spider Optimization
StArt	State of the Art through Systematic Review
SWRO	RO plant for sea water desalination
THRM	Thermal Energy
TVC	Thermal Vapor Compression
WT	Wind Turbine

the salt. Some of the RO advantages are:

- low energy consumption: the theoretical value is 0.05–1.0 kWh/m^3 for brackish water, recovery ratio of 70–75% and 1.06 kWh/m^3 for seawater, recovery ratio of 50% [18];
- membrane technologies evolution, aiming a higher recovery rate [19,20].

Osmosis, osmotic equilibrium and RO processes are shown in Fig. 1. A possible configuration of a solar/wind powered RO plant is shown in Fig. 2, highlighting three blocks: generation, storage and RO plant. The electricity to power the RO plant comes from photovoltaic (PV) modules and wind turbines (WT); batteries [22] and hydrogen (H_2) are options for energy storage, increasing the system reliability. The RO plant, for sea or brackish water desalination, is composed by a pump motor set, pre-treatment system, membrane, concentrate and drinking water tanks. Both the generation and storage blocks can supply the RO plant, depending on the solar and wind resource availability and the system management and control [23].

In such context, in this paper the combined use of Portfolio Theory (PT) and Particle Swarm Optimization (PSO) is proposed for the energy management of a Hybrid Renewable Energy System (HRES) powered RO

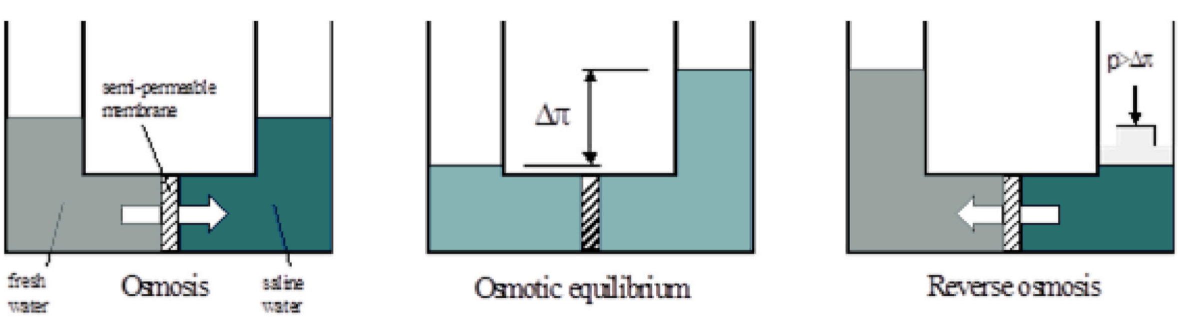

Fig. 1. Osmosis, osmotic equilibrium and RO [21].

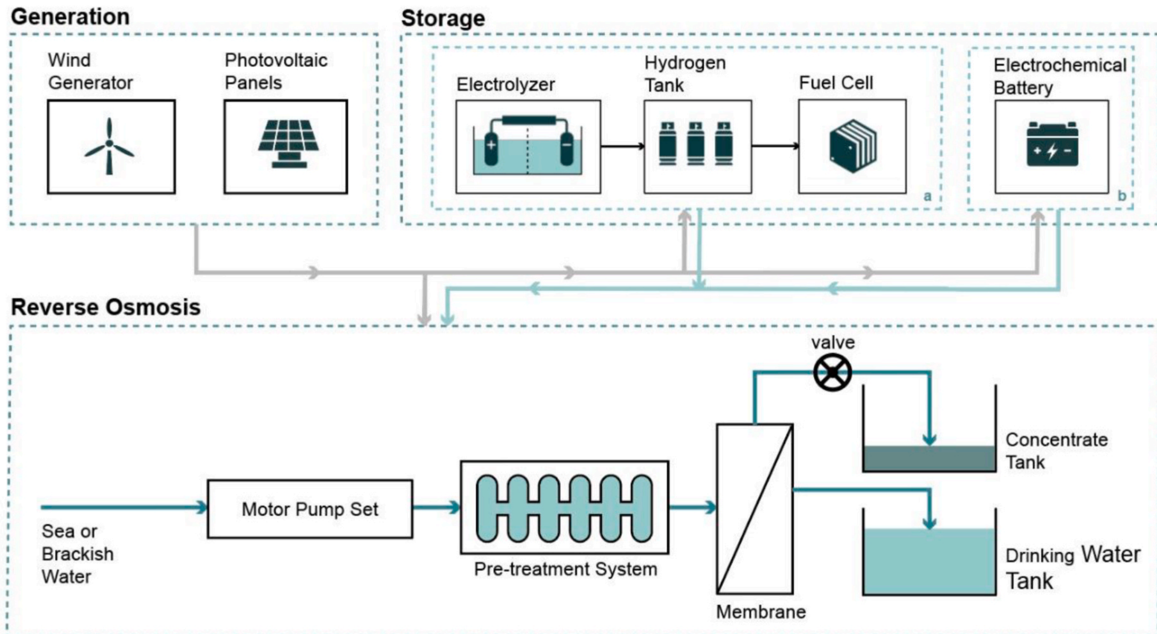

Fig. 2. Schematic diagram of a solar/wind powered RO plant.

plant. PT-PSO combined use aims a Multi-Objective Optimization (MOO), decreasing the Drinking Water Cost (DWC) and increasing the system reliability and components lifetime. Initially, a systematic and bibliometric review is conduced to identify the strengths and weakness of the existing HRES optimization methodologies. Optimization methods based on PSO show a lower computational cost, delivering robust results. Additionally, PSO can be used with other methodologies, such as PT; PT use for HRES energy management is innovative. The paper is divided in five sections: section one is the introduction; section two focuses on the bibliometric analysis and systematic review, section three presents the state-of-the-art of HRES powered RO plants optimization, in section four is proposed an optimization methodology for HRES powered RO plants and section five brings the conclusions.

2. Bibliometric analysis and systematic review

A systematic and bibliometric review is conducted to identify the strengths and weaknesses of the existing HRES optimization methodologies. The bibliometric analysis aims to identify trends of the research topic. The support of computational tools such as the State of the Art through Systematic Review (StArt) and VOSviewer is essential in this phase. The methodology flowchart is shown in Fig. 3.

The search is developed using Scopus, one of the best known and most renowned databases for abstract and citation. The questions used

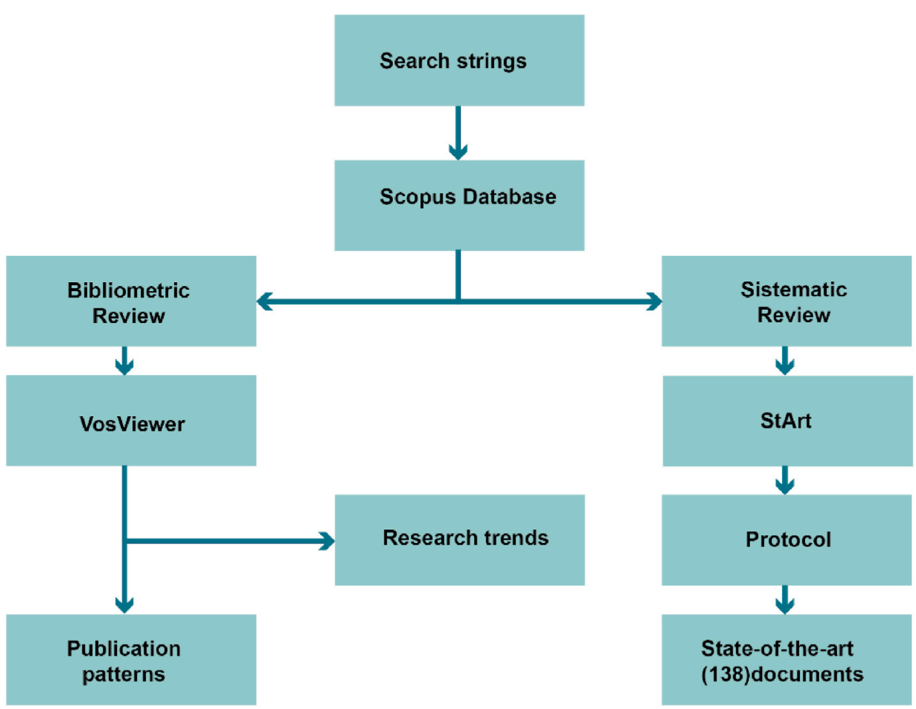

Fig. 3. Methodology flowchart.

to guide the review are formulated to characterize how to optimize HRES powered RO plants; the main question is: "What are the methodologies applied for the optimization of hybrid renewable systems applied to RO plants?". Using the platform, the first steps aim to select strings useful for the search: TITLE-ABS-KEY ((optimization AND renewable AND hybrid AND system AND to AND reverse AND osmosis) OR ((optimization) AND (renewable OR "solar and wind" OR "hybrid system") AND ("reverse osmosis" OR desalination)) OR (optimization AND "Hybrid System" AND "Reverse osmosis" AND storage) OR (metaheuristic AND "hybrid renewable system" AND "reverse osmosis")) AND PUBYEAR >2011.

The bibliometric parameters used to analyze publications on the research topic in the last ten years are:

(i) Type and language of documents;
(ii) Publication characteristics (authors, average number of authors per article);
(iii) Author performance;
(iv) Journal publishing patterns (number of publications, publication patterns by countries), journal – level metrics; collaboration and co-analysis citation, using Scopus © records and
(v) HRES search obtained by author keywords and Scopus keywords.

StArt is used after obtaining the articles with the search platforms, allowing the systematic review based on the previously defined protocol. A filter is applied to the articles, using title, authors, abstract and keywords. The inclusion and exclusion criteria are established in the protocol. At this stage, the database selected using Scopus offers a total of 374 documents to be analyzed; from these files, 60% of them are rejected based on the tittle and abstract, 2% are duplicated and 38% are selected.

Sequentially, the following steps are used in the Systematic Review to reduce the number of articles that remained after the first selection.

(i) Insertion of the articles data from Scopus in the BiBTex and RIS format or manual insertion;
(ii) Removal of duplicated articles and invalid results;

(iii) Enter of the inclusion/exclusion criteria for the title, abstract and keywords, the reading priority.

(iv) Papers accepted in the previous step are evaluated regarding the inclusion/exclusion criteria;
(v) Analyses of the results of the accepted articles.

Articles classification, according to the reading priority resulting from the application of these steps, is shown in Fig. 4.

VOSviewer is used to construct and visualize bibliometric maps. The tool application differential is its focus on the graphical representation of bibliometric maps, including journals, researchers, or individual publications, based on citation, bibliographic coupling, co-citation, or co-authorship relations [24].

2.1. Type of document, language of publications

The total of 374 documents can be classified as: Research Articles (70.3%), Conference Papers (13.6%), Reviews (7.8%), Book Chapters (4.5%), Conference Reviews (3.2%) and Books (0.5%), as shown in Fig. 5.

Journal and conferences papers are chosen to be analyzed for the creation of maps, according to Scopus and VOSviewer bibliometric analysis. StArt is used for the literature review of the articles with a direct connection with the theme. Focusing on the language, 368 papers are written in English, 5 in Chinese and 1 in Spanish.

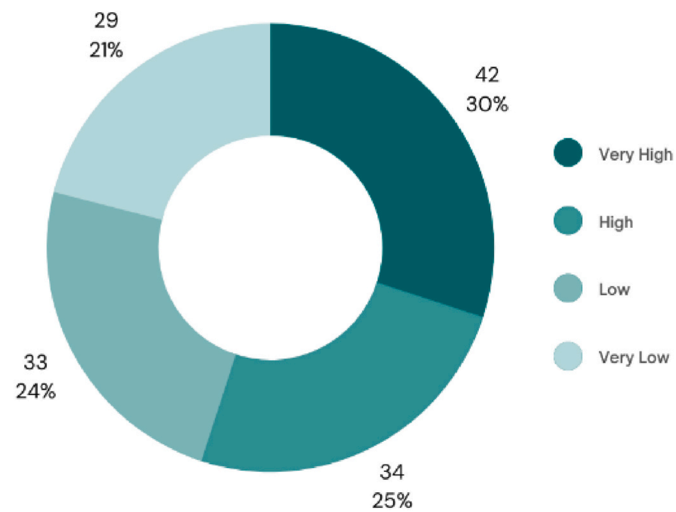

Fig. 4. Papers classification according to the reading priority.

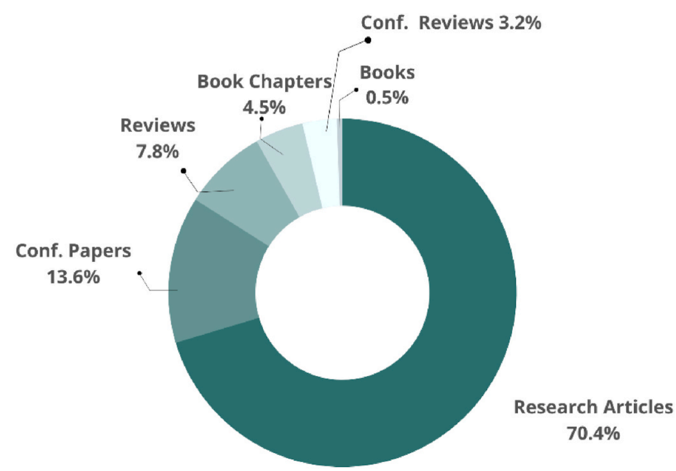

Fig. 5. Classification of the documents (Scopus database).

2.2. Publications characteristics, author, and country performance

The number of publications on HRES powered RO plants from 2012 to May 2022 is shown in Fig. 6. According to the data, this number has grown considerably over the years, highlighting a peak of 76 papers in 2021; before this year, the highest number is registered in 2019, with 47 publications. Until May 2022, almost 30 publications is found, confirming the increasing tendency.

The countries with the highest number of publications on HRES

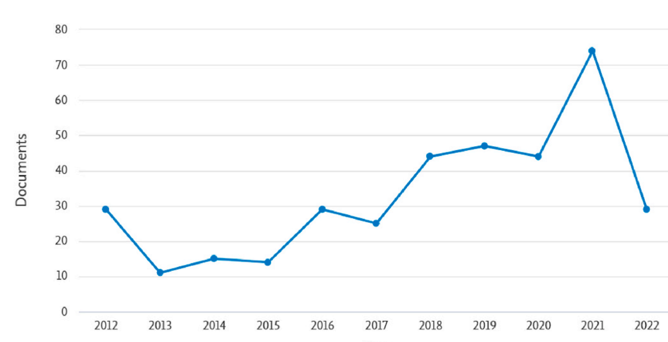

Fig. 6. Number of publications on HRES powered RO plants, 2012–May 2022 (Scopus database).

powered RO plants are China (55), Iran (47), USA (43), Egypt (32), Saudi Arabia (28), United Kingdom (24) and Spain (23), as shown in Fig. 7.

The number of Chinese papers can be associated with the increasing water scarcity in some areas of the country, especially in the northeast region. This process is due to the rapid economic growth, poor water spatial distribution and population growth [25]; most of the papers focus on RO plants for sea water desalination (SWRO) [26,27].

2.3. Journal publishing standards and scopus subject category

From the documents found from the bibliometric analysis, 126 are published in 6 different journals; the number of papers per journal from 2012 to May 2022 is shown in Fig. 8. "Desalination" concentrates most of the publications (45), followed by "Energy" (23), "Desalination and Water Treatment" (17), "Energy Conversion and Management" (14), "Renewable Energy" (14) and "Applied Energy" (13).

2.4. Co-autorship

VOSviewer based bibliometric analysis provides a graphical representation of the collaboration network; in Fig. 9 are shown the top authors of the research area, identifying a total of 16 collaborations between them.

In Fig. 10 are shown the collaboration between countries. China, the country with most of the publications, focuses on the collaboration with USA and Egypt; Iran focuses on United Kingdom, United Arab Emirates and Spain.

2.5. Author keywords and indexed keywords

An analysis of trends is implemented based on the authors keywords and the indexed keywords. Authors keywords are chosen by the authors, representing the survey content, while indexed keywords are chosen by Scopus, based on Elsevier's Thesauri, considering synonyms and other grammatical characteristics [[28]]. Indexing is important for the elaboration of indexes, recompiling and ordering the data, aiming to trace a relationship between them. Author keywords are classified for frequencies $\geq$5, providing a total of 25 keywords, as shown in Fig. 11. The highlighted keyword is "desalination", with a total of 105 occurrences and a binding force of 80, representing the number of associations of the keyword with other descriptors.

Indexed keywords are classified for frequencies $\geq$10, providing a total of 117 keywords, as shown in Fig. 12. The highlighted keyword is "desalination", totaling 279 occurrences and a binding force of 277. "Optimization" (143 occurrences) and "reverse osmosis" (113 occurrences) are the second and third keywords with the highest number of occurrences, showing binding forces of 142 and 113, respectively. Sequentially, "renewable energy sources" comes associated with the RO

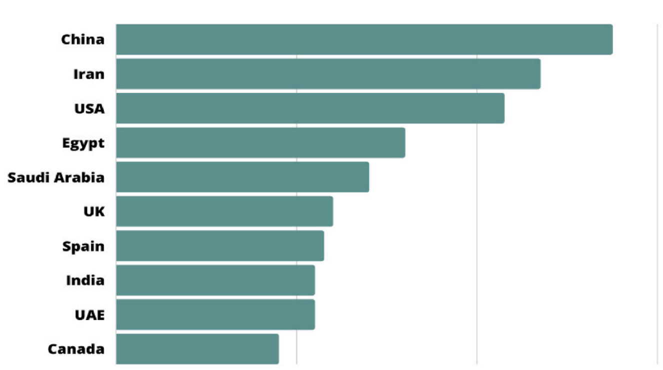

Fig. 7. Number of publications on HRES powered RO plants per country (Scopus database).

process to produce potable water. "Hybrid system" is directly connected with "optimization", highlighting the solar and wind power intermittence.

3. Optimization of HRES powered RO plants

The reduction of the energy consumption of desalination processes is a necessary step as the plant capacity grows [29]. One of the RO process main parameters is the specific energy consumption (SEC) of the associated drinking water production, significantly influenced by the process efficiency [30]. In such context, a PV/WT powered 100–1000 m^3/day RO unit, with a Battery Storage System (BSS) and control unit, is optimized in Ref. [31] aiming the minimum SEC, based on the plant productivity, interest rate and lifetime. A technical and economical optimization is proposed for a PV/WT powered RO plant to minimize the DWC per m^3 [32]; the plant is tested with the generators operating together and separately; the results are compared with Organic Rankine Cycle (ORC) powered RO plants. Hence, most of the researches aims to minimize the DWC and SEC, increasing the reliability [33,34] and focusing on PV/WT powered RO unities [35]. An optimal function is applied to a PV/WT powered RO plant in Huludao, China [36]; a pump system is used to pump water to a reservoir and to a RO unity; DWC is minimized using a mixed-integer linear programming (MILP). Nuclear power and WT are applied to Freeze desalination (FD) and RO [37]; RO shows to be more cost effective. MILP is used to the design and sizing optimization of a grid connected PV/WT powered RO plant in Saudi Arabia [38]; a minimum DWC between 1.72 and 1.84 US\$/m^3 is found, with a SEC between 2 and 4 kWh/m^3. A PV/WT system to power a hybridization of two desalinations methods, RO and Multi-stage flash (MSF), is analyzed in Ref. [39], highlighting optimization and sensitivity analysis for different models and costs, showing an optimal DWC between 1.35 and 1.84 US\$/m^3. Python toolbox stochastic optimization is applied to a PV/WT plant with a BSS in Ref. [40], aiming to minimize the costs. An energy system optimal management is investigated in Ref. [41] to reduce the PV/WT critical excess of electricity production to power a RO unit in Jordan. A PV/WT plant with H_2 storage to power a RO unit in Iran is studied in Ref. [42]; the optimization is based on the Life Cycle Cost (LCC) and Loss of Power Supply Probability (LPSP), aiming to increase the reliability. Different RO unities using PV/WT/Diesel Generator (DG) and BSS are compared in Ref. [43]; optimization is applied to improve the reliability, considering HRES for remote communities. The most relevant data about the mentioned plants are summarized in Table 1.

3.1. HRES powered RO plants with energy storage

Renewable energy intermittence can cause fluctuations for the operation of RO plants without energy storage, causing hydrodynamic changes and affecting the driving force [88]. Considering such restriction, a division algorithm (DA) is applied to a 1.5 m^3/h SWRO [86], considering three different combinations: PV/WT/BSS/SWRO, PV/BSS/SWRO and WT/BSS/SWRO; the first plant shows minimal DWC and environmental damage, increasing reliability to remote regions. An improved bee algorithm is used to optimize a PV/WT powered RO plant [50]; six configurations are considered: PV/BSS/RO; PV/H_2/RO; WT/BSS/RO; WT/H_2/RO; PV/WT/BSS/RO; PV/WT/H_2/RO.

An algorithm is developed based on load demand, HRES production, battery and inverter [45], to study RO plants techno-economic aspects and meteorological specifications in Egypt. 150 m^3/day and 300 m^3/day RO plants are investigated, with a SEC of 7.3 kWh/m^3 and 4.6 kWh/m^3 and DWC of 0.463 US\$/m^3 and 0.376 US\$/m^3, respectively. An annealing-chaotic search algorithm is applied to three different system arrangements [51]: PV/WT/BSS/RO; PV/BSS/RO and WT/BSS/RO; the algorithm improves system performance and reliability, meeting energy demand and lowering costs. Numerical simulations of combined PV/WT/BSS and RO plant for the water supply of the Caribbean islands

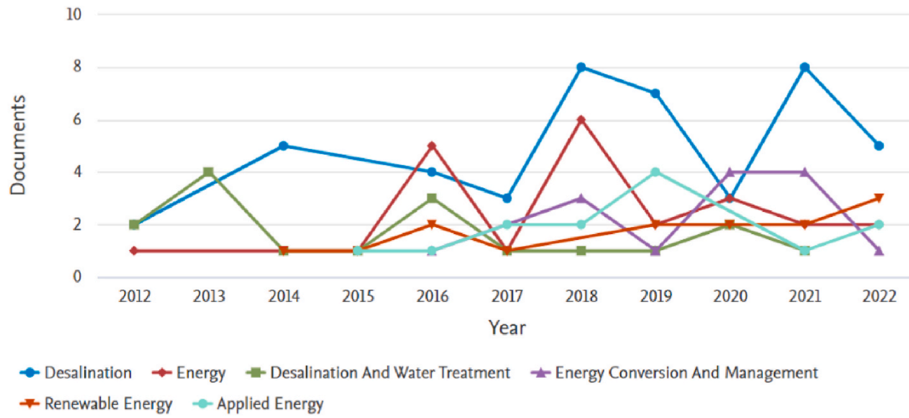

Fig. 8. Number of papers per journal from 2012 to May 2022 (Scopus database).

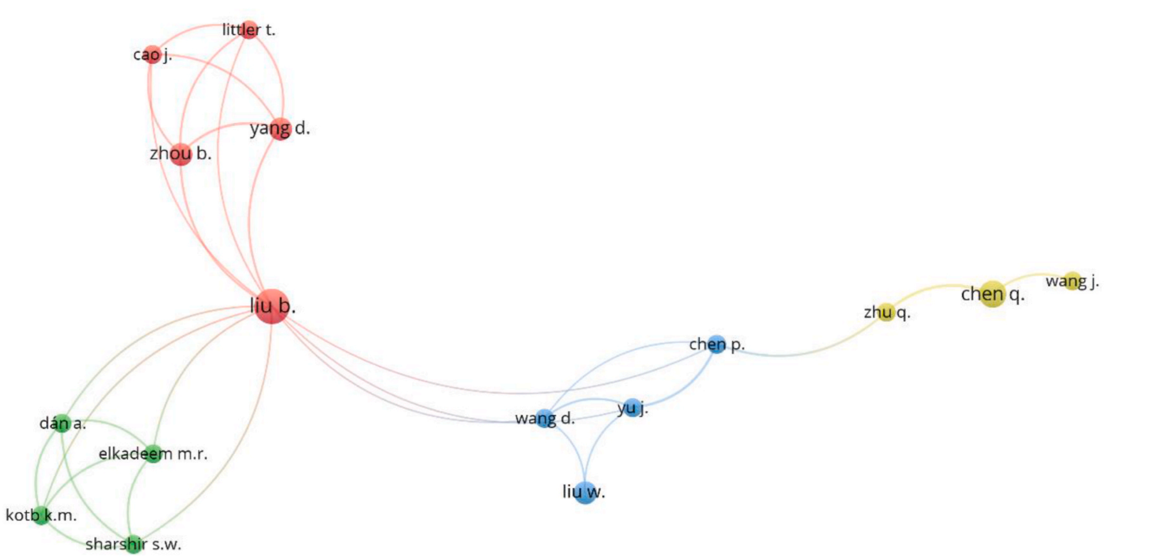

Fig. 9. Collaboration network – co-authoring (VOSviewer and Scopus Database).

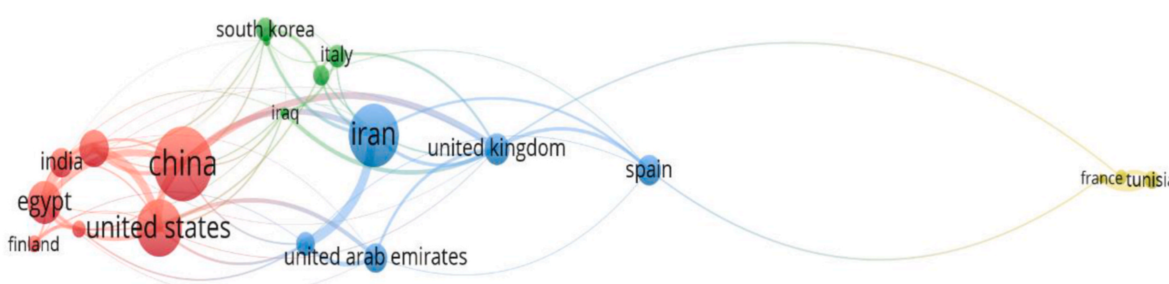

Fig. 10. Collaboration between countries (VOSviewer and Scopus Database).

are proposed [44].

A HRES with and without H_2 storage to power an RO plant are compared [89], analyzing costs for various scenarios. The feasibility study considers the possible replacement of fossil fuel-based power systems in Masawa, Eritrea and Newhaven, Southeast England to supply around 50,000 people. An optimization of an autonomous system with different configurations of PV/biomass/BSS/DG/RO for the energy and water supply of a village in India is proposed [75]. Twelve types of batteries are combined in different system configurations using algorithms, aiming an optimization; salp swarm algorithm (SSA) shows a better convergence and robustness.

A bee swarm optimization (BSO) is applied to six different configurations; sequentially, the results are compared with harmony search algorithm optimization (HSAO), showing that BSS is more cost effective than H_2. A PV/WT/BSS powered 200,000 m^3/day RO unit to supply drinking water for about 550,000 people in Algeria is analyzed [57]; an optimization model is developed to increase the system reliability. A power pinch analysis optimization is proposed in Ref. [55] to minimize the power losses and the external dependence on potable water. The system consists of a PV/WT/BSS and RO plant. Three different

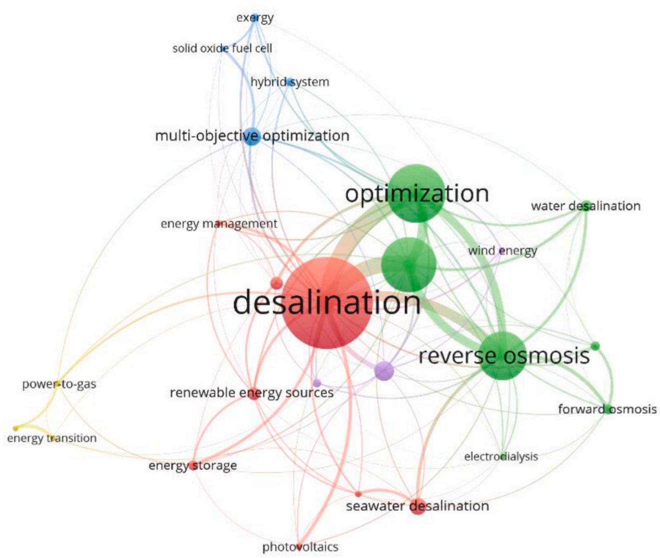

Fig. 11. Author keywords (VOSviewer).

configurations are investigated: PV/WT/BSS, PV/WT/DG and PV/WT/BSS/DG [90]; an optimization aims to minimize the total annual and CO_2 emissions cost. A genetic algorithm (GA) is used to optimize a PV/WT/BSS/RO system [47]; a minimization objective function is applied to find the best DWC.

A cost optimization is developed for a PV/WT/DG/BSS microgrid system [73] to power a 100 m^3/day RO plant in Egypt, considering eleven different scenarios. DA is used to optimize a PV/WT/DG/BSS/SWRO plant in Iran [91]. A multi-objective optimization is applied to minimize the total LCC and environmental impact. BSO and GA, are compared; DA is faster than the first two. GA and PSO, multi-objective evolutionary algorithms, are applied to a standalone PV/WT/DG/BSS/RO [87]; the excess of energy is used to provide reliable electricity and potable water supply to a community in Bangladesh. GA and PSO combination decrease the overall system cost (11.8%) and electricity cost (9.3%), increasing the community employments (6%).

3.2. Multi-objective optimization

A new model aiming to optimize a PV/WT powered RO unit, designed to supply the water demand of a small community in the Emirate of Abu Dhabi, is proposed in Ref. [46]. MOO is used to obtain the lowest DWC; three SEC values are considered: 2.5, 5.0 and 7.5 kWh/m^3. A Multi-objective Multi-verse Optimization (MOMVO) is applied to a PV/WT/BSS/DG micro-grid [92], to power a RO plant and supply a residential load. Two scenarios are analyzed: a) only renewable sources and b) use of a DG. The optimization focusses on the cost of electricity (COE), LPSP and Renewable Factor (RF). A grey wolf optimization (GWO) algorithm is applied to solve a multi-objective problem [93], aiming to minimize operating costs and net load fluctuation. A PV/WT/BSS system powers a RO plant and electric vehicles, in four different scenarios. A PV/WT/BSS system is integrated to a micro grid to provide potable water, cooling, power and heating [94]; MOO seeks to minimize LCC and CO_2 emissions, using GA.

A cogeneration system, parabolic trough collector (PTC) and WT, is optimized for electricity and potable water supply [70]. Three different desalination technologies are studied, including RO; a multi-objective PSO is applied. The results show an exergy efficiency of 26.2% and a DWC of 3.08 US$/$m^3$. When RO operates with a Multi Effect Distillation (MED) system, an energy economy of 8.9% is found. A MOO minimizes net present cost (NPC) and increases renewable energy proportion based on the Pareto's frontier [81]. The system is composed by PV/WT/BSS/H_2 and a RO plant to wastewater treatment. A PV/WT plant combined with other sources, including natural gas (NG) and coal gasification, is investigated to power a large-scale RO unit [54]; MOO minimizes the COE and CO_2 emissions. A multi-objective strategy is applied to a modular reactor and WT/BSS/RO plant in Ref. [59] to find

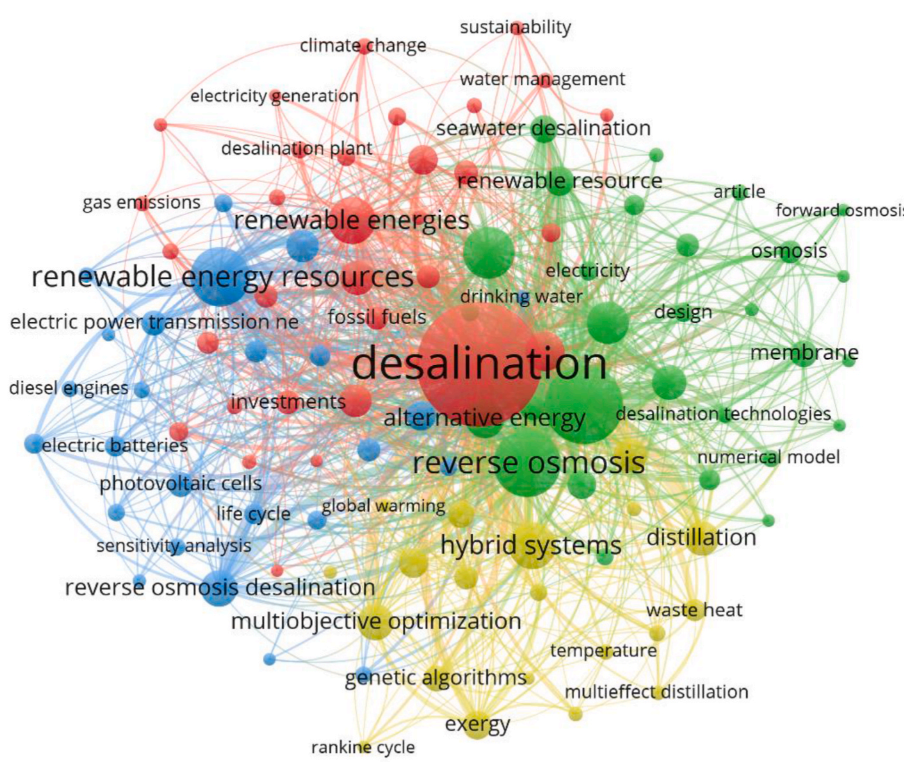

Fig. 12. Indexed keywords source: VOSviewer and scopus database (2022).

Table 1
Relevant data on HRES powered RO plants.

Ref	Year/Site	Production (m^3/day)	SEC (kWh/m^3)	DWC (US $/m^3$)	Configuration/ Load
[44]	Grenada (2012)	1.5×10^2	11	1.91	PV/WT/BSS/ DG/RO/LD
[45]	Egypt (2012)	1.5×10^2–3×10^2	7.3–4.6	0.46–0.37	PV/WT/BSS/ RO/LD
[46]	UAE (2013)	$1,5 \times 10^3$	2.5–5–7.5	0.49, 0.85, 1.21	PV/WT/RO
[47]	Tunisia (2013)	15	N/A	2.62	PV/WT/BSS/ RO
[48]	Saudi Arabia (2014)	N/A	4.5–1.0 - 4.0	0.47, 0.54, 0.65	PV/WT/RO/ ODMs
[49]	Honduras, Eritrea, Australia (2015)	10	4	3	PV/WT/BSS/ DG/RO
[42]	Iran (2016)	10	4	N/A	PV/WT/H_2/ RO
[50]	Iran (2018)	10	4	N/A	PV/WT/H_2/ RO
[51]	Iran (2018)	10	4	255.4	PV/WT/RO
[41]	Jordan (2016)	1.55×10^3	2.72–4.75	N/A	WT/PV/RO/ LD
[52]	Iran (2016)	10	4	N/A	WT/PV/RO
[53]	Egypt (2016)	2	N/A	N/A	WT/PV/DG/ RO
[54]	Egypt (2016)	3×10^3	3–4	N/A	PV/WT/NG/ CG/CHP/RO/ LD
[55]	UK (2017)	960	3	0.11	PV/WT/RO
[56]	Egypt (2018)	N/A	3–4	1.30–1.52	WT/PV/DG/ RO
[40]	Spain (2018)	$2,8 \times 10^3$	N/A	N/A	PV/WT/RO
[57]	Algeria (2018)	2×10^5	0.21	N/A	PV/WT/RO
[39]	Iran (2018)	N/A	N/A	1.35–1.84	PV/WT/RO/ MSF
[58]	Turkey (2018)	24	4.38	2.20	WT/PV/DG/ RO
[59]	USA (2018)	N/A	1.4	0.74	Nuclear/WT/ RO
[60]	Egypt (2018)	3×10^2	N/A	N/A	WT/PV/DG/ RO
[61]	Iran (2018)	10	4	255.4	PV/WT/RO
[62]	Greece (2018)	11.5–11.8	N/A	N/A	PV/WT/RO
[63]	Italy (2019)	1.6×10^3	6	7.13–10.19	WT/PV/DG/ RO/LD
[31]	Egypt (2019)	1×10^2 – 10×10^3	3.5–6.7	0.86; 0.93; 1.25	PV/WT/RO
[64]	Lebanon (2019)	1×10^3- 5×10^4	2–5	6.95–24.29	PV/CSP/WT/ GTE/NG/FO/ RO/LD
[65]	Portugal (2019)	3.3×10^5	3	74.02	PV/WT/RO/ LD
[66]	USA (2019)	18.93	5	3.84–4.48	PV/WT/DG/ RO
[67]	Spain (2019)	50	5	N/A	PV/WT/DG/ RO
[68]	Spain (2019)	5.6×10^3	3	0.03	PV/WT/RO/ PG
[69]	South Africa (2020)	3.5×10^4	3-3.7–4.5	1056	PV/WT/RO/ BTU/ED
[70]	Iran (2020)	2.3×10^4	7.71	3.08	PTC/ORC/ WT/RO/TVC
[71]	Iran (2020)	3.15×10^5	N/A	532	PV/WT/tidal/ RO/LD
[72]	Spain (2020)	4.8×10^3	1.5-3-10-14	N/A	PV/WT/DG/ LD/RO/THRM
[73]	Egypt (2020)	1×10^2	507.32	N/A	PV/WT/DG/ BSS/RO
[74]	Egypt (2020)	1.5×10^3	2.0–4.0	N/A	PV/WT/DG/ RO
[75]	India (2020)	5	2	N/A	PV/WT/BM/ RO/LD
[76]	Egypt (2020)	8.5×10^2	N/A	N/A	PV/WT/DG/ CHP/LD
[77]	USA (2020)	18.93	N/A	3.35	PV/WT/BSS/ DG/RO
[78]	Greece (2021)	1×10^2–2×10^3	N/A	0.47–1.38	PV/WT/RO
[79]	Oman (2021)	10	5.8	1.74	PV/WT/DG/ RO
[32]	Sues Gulf (2021)	>3×10^3	7.68	0.55; 0.53; 0.57	PV/WT/CSP/ RO
[80]	Egypt (2021)	1×10^3	N/A	N/A	PV/WT/DG/ RO
[81]	Italy (2021)	41.6	0.51	2.5–23	PV/WT/RO
[82]	Jordan (2021)	1×10^2	N/A	N/A	PV/WT/DG/ RO
[83]	Saudi Arabia (2021)	1×10^3	2.44	0.75	PV/WT/RO
[38]	Saudi Arabia (2021)	8×10^2	2–4	1.72–1.84	PV/WT/RO
[84]	Tunisia (2021)	60	0.5–2.5	N/A	PV/WT/RO
[85]	Tunisia (2021)	28.50	N/A	N/A	PV/WT/RO
[86]	Iran (2021)	36	4.27	647	PV/WT/SWRO
[87]	Bangladesh (2022)	20	4.38	532	PV/WT/DG/ RO/LD
[75]	India (2022)	30	3	4.57	PV/WT/BSS/ DG/RO

*EUR 1 = US$ 1.0245–19 July 20, 2022.

the levelized cost of electricity (LCOE), using stochastic optimization and a parametric analysis, increasing the reliability.

3.3. *Multi-energy systems (MES)*

Several countries are investing in a 100% renewable energy scenario. Hence, the combination of various renewable sources can be highly effective in such approach [95], supplying cooling, heating, electricity and drinking water. Although most of the MES include not only renewable sources, models try to reach a high percentage of renewable energy converters; optimization is a very important step in such process [96–98], dealing with the interactions between energy sectors [99]. An energy management approach is applied to electricity and drinking water production [48], which is stored in case of excess of renewable energy; the RO plant is operated as deferrable load, improving the reliability. A robust optimization method is applied to a MES aiming to provide drinking water, electricity, heating and cooling [100]; a PV/WT system powers a RO plant, a DG is added as backup. A MES provides drinking water, heating, cooling and power for isolated buildings, using WT/PV/BSS and liquefied petroleum gas, a low carbon fuel [101]; potable water is provided by a RO plant. An Artificial Neural Network (ANN) estimates the energy consumption of a RO plant; integrating ANN with PSO, the RO plant electricity consumption is reduced about 4,95%, resulting in 733 kWh/year. An optimization of a PV/WT/DG/BSS/RO is developed in Southern India [102]. The optimization considers economic and environmental impacts, finding a minimum DWC of 4.57 US $/m^3$ and a minimum CO_2 production of 2887 kg.

An energy hub is composed by three units: a) PV/WT to provide cooling, heat and power; b) a storage energy system and c) a flexible

SWRO plant [103]. The optimization aims to minimize environmental and operation costs, considering the sources intermittence and other demands. The optimization of a PV/WT/BSS/DG powered large-scale RO unit in Egypt, providing heat, electricity and potable water, is proposed [76], aiming to minimize emissions, NPC and energy cost. Different renewable resources are investigated for a 100% renewable energy scenario in the Philippines [104], providing energy to RO and MED desalination, heating, transportation and power. A 100% renewable MES, based on power-to-gas and biogas technologies, is optimized [105] to provide heating, cooling, gas and potable water to an island; a MOO is proposed considering extremes weather conditions. Power-to-gas is used as energy storage, showing to be more cost effective than batteries storage. A PV/WT/SWRO, using the concept of distributed generation for the PV/WT system, is optimized under different operating models, using linear complex algorithm [106].

3.4. Artificial intelligence (AI)

The use of AI can help in the case of complex nonlinear problems, difficult to solve with simple mathematical models. Considering HRES powered RO plants, AI is usually used to optimization, control by sequence, prediction and expert decision making; GA is one of the most used algorithms for the optimization of such systems [107]. An optimization methodology is applied to a PV/WT/BSS/RO in Southern Tunisia [84]; the non-dominated sorting genetic algorithm-II (NSGA-II) is based on the minimization of two objective functions: a) the embodied energy and b) the reliability indicator. A PV/WT/BSS system powers a 100 m^3/day RO unit, in Saudi Arabia [83]; three optimization algorithms are applied to minimize costs and LPSP: GA, PSO and Bat Algorithm Optimization (BAO); BAO shows a cost of 0.745 US$/$m^3$ and a reliability of 99.4%. PSO is used for the technical economical optimization of a PV/WT powered RO plant [78]. Twenty-five on-grid and off-grid configurations are tested, with RO plants ranging from 100 to 2000 m^3/day; the results show that the addition of RE is more cost-effective, even to grid-connected systems.

A multi-criteria optimization based on GA is applied to the sizing of a PV/WT/RO plant in Tunisia [108]. The optimization algorithm allows to find the best arrangement of PV arrays, WT and water storage tanks capacity, resulting in a more cost-effective system. The optimization of a HRES powered RO plant is proposed for the Southern Tunisia [85], applied to two objective functions, using bi-objective GA: a) reliability indicator and b) environmental indicator. PSO, BSO, Harmony Search Algorithm Optimization (HSAO), Cuckoo Search (CS), Search Algorithm Optimization (SAO) and Tabu Search are applied to optimize a PV/WT/BSS/RO system in a remote area in Iran [61]. The algorithms are applied to minimize the LCC and increase the reliability. An off-grid PV/WT/DG/BSS powered 18.93 m^3/day RO unit is optimized [109], using an Ant system (AS) to find the system optimal configuration; DWC decreases almost 50%, compared with the actual price. A Hybrid Chaotic Search/Simulated Annealing Algorithm (HCS/SAA) is applied to minimize LCC and LPSP of a PV/WT/RO system [109]. A PSO-GWO is used in Ref. [56] to optimize an on-grid PV/WT/BSS/H_2/DG powered RO unit, minimizing DWC and CO_2 emissions; the best results are obtained using batteries instead of hydrogen.

3.5. Software for HRES optimization

A PV/WT/BSS/DG powered 1 m^3/h RO unit in Turkey, with a cost of 0.308 US$/kWh is investigated [58]. The Hybrid Optimization Model for Multiple Energy Resources (HOMER) and Reverse Osmosis System Analysis (ROSA) are used to optimize the HRES and to analyze the RO model, respectively. HOMER is also applied to the optimization of a PV/WT/DG powered 100 m^3/day RO unit in Jordan [82], focusing on NPC and COE and showing different system configurations; the best alternative is a PV/WT/DG with a DC/AC converter. A PV/WT/DG/RO, with a storage coupling pumped-hydro storage and flow-battery, is analyzed in Egypt [80]; HOMER is used to optimize NPC and COE; FUZZY is used for the decision making process. A PV/WT/BSS powered RO/electrodialysis (ED) unit is investigated [110]; the optimization aims to minimize the water and electricity consumption. HOMER is used for the design and optimization of a standalone PV/WT/RO plant in the Canary Islands [111]. The optimization of a PV/WT/BSS/RO is compared with a PV/WT/BSS/RO using DG as backup [79]. The HRES electricity cost is about 0.37 US$/kWh; the cost with DG is 0.28 US$/kWh.

The design and optimization of a PV/WT/BSS powered 50 m^3/day RO plant in the Canary Islands using HOMER is commented in Ref. [67]. The results show that AI could be more effective in different geographic areas due to the parameters variation. HOMER is also used for the design and optimization of a PV/WT/BSS/DG powered 1500 m^3/day RO unit [74], aiming to find the best configuration, providing the lowest COE (in this case, 0.107 US$/kWh). Two configurations are chosen: with and without batteries, providing eleven different systems. A PV/WT/BSS/DG system is designed and optimized using HOMER, to provide water and energy supply to the island of Lanzarote and Fuerteventura [72]. The PV/WT system is responsible to supply electricity and thermal energy, DG is used as backup; various desalination processes are considered: RO, MSF and MED; RO shows the best LCOE and renewable energy penetration. A Social Spider Optimization (SSO) is applied to a PV/WT/BSS/Tidal/RO [71], calculating LCC and increasing the water availability; the power loss is reduced in 2%. HOMER is used for the optimization of a PV/WT/RO [53], with and without DG, based on the NPC and COE for a small greenhouse hydroponic cultivation. An off-grid PV/WT/DG/BSS powered 300 m^3/day RO unit is studied [60]. HOMER is applied for the sizing and economical optimization; the best COE is 0.282 US$/kWh for a 186 kW PV plant, eight WT of 100 kW and a 350 kW DG.

3.6. Levelized cost of energy and water

The optimal sizing and energy-efficient management of a HRES and electrical grid is developed in Ref. [69] to power RO and ED plants and crystallization methods. The goal is to increase the drinking water production, reduce the brine production and the LCOE. A PV/WT/BSS is used to power a 5600 m^3/day RO unit and to supply electricity to the grid, reaching 5.88 GWh/year, in the Canary Islands [68]. The optimization is based on the “black stochastic optimization box”, showing a LCOE minimization. A PV/WT/BSS/DG powers a 3.95 kW RO plant, with a production of 18.93 m^3/day [66]. GA is used to solve the optimization problem and to find the minimum levelized cost of water (LCOW).

A PV/WT powered 330,000 m^3/day SWRO in Portugal is investigated [65]. The electricity cost minimization, considering CAPEX and OPEX, can reduce the LCOW. A MILP is applied for the optimization of various energy systems, attending water and energy demand. SWRO, RO plant for brackish water desalination (BWRO) and ED are compared, using PV/WT, Concentrated Solar Power (CSP), NG and others [64]. The system is connected to the grid and the objective function is the minimum annual cost. A multicriteria optimization is applied to a PV/WT/RO to minimize the total annual costs and emissions, improving energy reliability [112]. Considering the forecasting of the sources behavior and of the potable water demand, the potential loss of power supply probability (PLPSP) is 18.3% less than the base case (no forecasting). A novel dynamic MOO is applied to the design and operation of an autonomous PV/WT/DG system to provide water and electricity for Ustica island, in Italy. The RO unit has a production of 1600 m^3/day; the objective functions to be optimized are NPC, water deficit and electricity surplus [63]. A power management optimization, using fuzzy logic and GA is applied to an off-grid PV/WT/BWRO system with hydraulic storage. The goal is to increase the drinking water production and to optimize the power sharing factor, based on the renewable energy distribution and different energy services [62]. Strengths and weakness of the existing HRES optimization methodologies are shown in Table 2.

4. Proposed methodology for HRES powered RO plants

Based on the systematic and bibliometric review, the combined use of PT and PSO is proposed for the energy management of a HRES powered RO plant. PT - PSO combined use for the energy management of a HRES powered RO plant can provide efficient and robust results, compared to others optimization algorithms. Considering that PSO is based on a global optimization, a significant number of possible

Table 2
Strengths and weakness of HRES optimization methodologies.

Methodology		Reference	Pros	Cons
Commer-cial Software	HOMER	[1,21,23,35,43–45,47,52,57,67]	Modeling flexibility; technical and financial feasibility, user-friendly interface.	Paid software, need of technical knowledge, dependency of the input data quality, limited range of the components nominal power.
	TRNSYS	[113]	Wide range of components, accurate simulation results, customizable optimization, various analysis options.	Paid software, Steep learning curve, time-consuming simulations, limited flexibility in optimization methods.
AI	GA	[47,66,84,94]	Accurate and efficient optimization; convergence to global optimum solution, computational simplicity.	Slow convergence, sensitivity to parameters, difficulty finding optimal solutions, need of prior knowledge of the problem.
	PSO	[101]	Easy implementation, fast convergence depending on the dimensional search space, no use of gradient, versatile.	Slow convergence to reach the global optimum in high dimensional search space, premature convergence, sensitivity to parameters, lack of theoretical guarantees, limited scalability.
	GWO	[93]	Effectiveness, flexibility, speed.	Local optimal convergence, sensitivity to initial values, limited applicability.
	BAO	[83]	Robust and adaptive, rapid convergence to global optimal solutions, easy implementation.	Sensitivity to parameter values, potential convergence to suboptimal solutions, limited applicability.
	AS	[77]	Robustness; adaptability; scalability; global optimization; easy implementation.	Computational complexity; limited accuracy; lack of guarantees; sensitivity to parameter settings; difficulty in handling discrete variables.
	SSA	[75]	Efficient and robust for problems with rapid convergence to global optimal solutions, flexible and versatile.	Lack of established theory, sensitivity to parameters, limited applicability.
	MPSO	[90]	Efficient and robust optimization, flexible and versatile, outperforms traditional PSO methods.	Sensitivity to parameter values, limited applicability, complexity compared with traditional PSO.
	EMP /ANN	[112]	Accurate modeling; robustness; flexibility; reduced computation time; easy implementation; interconnectivity.	Lack of data availability; parameter tuning; overfitting.
HYBRID	HGAPSO	[87]	Increased solution quality, improved convergence, flexibility, ability to handle complex problems.	Requirement of fine-tuning of parameters, no guarantee of global optimal solution, computationally expensive.
	GenOpt, TRNSYS, PSO	[78]	Accurate modeling; robustness; flexibility; parallel processing; multiple objectives.	Computational complexity; difficulty in model integration; parameter tuning; limited application.
	RAVEN	[59]	Handling of noisy and non-smooth objective functions; efficient use of computational resources; good performance in high-dimensional problems.	Local optimal convergence; Requirement of careful selection of step size for good performance; Limited applicability.
	GA, Nelder–Mead simplex	[49]	Powerful optimization techniques for HRES.	Requirement of significant computational resources; no guarantee of global optimal solution.
	HSCSA	[61]	Handling of multiple objectives and noisy or incomplete data; efficient solution finding; easy implementation; reduced computation time.	Sensitivity to initial parameter settings; local optimal convergence; difficulty in interpreting trade-offs; requirement of high-quality data.
	SSO	[71]	Multiple optimal solutions; relatively easy implementation; computationally efficient, handling of complex and non-linear problems.	Convergence to suboptimal solutions, uncertainty and stochastic limited ability, limited research and application compared to other methods.
	MOMVO	[92]	Multi-objective optimization; Pareto optimal front; robustness to noisy or incomplete data; easy implementation; scalability.	Computationally intensive; local optimal convergence; data quality dependency; Interpretability challenge; sensitivity to initialization.
	MCDM	[102]	Handling of multiple objectives, transparency, flexibility, trade-off analysis, risk management.	Subjectivity, data requirements, complex calculations.
Mathema-tical Model	Sensitive analysis	[39]	Optimization of important factors; robustness in uncertain conditions; identification of potential failures; real-time optimization capabilities; easy implementation.	High computational requirements; sensitive to input variable selection; high-quality data for accuracy; no consideration of all performance factors; no suitability for all systems.
	MINLP	[110]	Discrete and continuous variables optimization; efficiency with multiple objectives; guarantee of global solutions; real-time optimization; handling of constraints.	Computationally intensive; limited software and expertise; requirement of high-quality data; no guarantee of optimal solution; results difficult to interpret.
	Technical, analytical optimiza-tion	[31,32]	Interpretable solutions; handling of constraints and variables; efficiency; no need of specialized software or expertise.	Only solvable models, no guarantee of global solutions, limitation to complex systems, requirement of mathematical knowledge.
	Load demand increase	[45]	Simple and cost-effective solution; improved system performance.	No suitability for large-scale systems with great variability of load demand; no consideration of financial constrains; no guarantee of global optimal solution.
	DA	[86,91]	Monitoring and control improvement; reliability and efficiency; outage times and costs reduction; better integration of renewable energy.	Requirement of significant investment and regulatory changes; need of specialized expertise; limited effectiveness in complex systems; vulnerability to cyber-attacks.
	Linear Program-ming		Efficient optimization; systematic decision-making approach; scenario analysis.	Only linear relationships are considered; limited effectiveness with some renewables; high computational requirements for large systems; limited ability with discrete variables.

combinations can be found, as all the search space is used to find the optimal solution, providing a range of possible feasible solutions.

PT offers the possibility to deal with uncertainties, associated with the different types of energy sources and storage options used in HRES. Such methodology can help to find the sources allocations with the minimum risks for the expected returns. In the area of solar resource forecasting, the integration PT - ANN gives more accurate results than using individually AI and PT [114]. PT was developed in 1950 by Harry Markowitz; since then, PT has been applied in the economy sector to improve investment portfolios, reducing investments risk and increasing the return, diversifying the type of investments. In the energy sector, PT has been applied to decrease the forecasting error of solar and wind resources [114–116]. Other PT applications in the sector are related to the energy market, including optimization of portfolios of renewable systems components, aiming to choose the best cost and increase reliability. PT analyses the risk and maximize the expected return of each asset, then combines these assets in a portfolio, minimizing the total risk. This optimization is based on a historic performance of the assets, finding the correlations between them and creating a balance between risk and return, offering the best options for investors.

PT-PSO combination can offer more flexibility to choose the objective functions and their parameters and constrains, offering the possibility of adaptation to many different HRES powered RO plants configurations. Additionally, such combination can help to solve multi-objective problems, considering different criteria at the same time, as energy cost, reliability and lifetime. The proposed methodology is shown in Fig. 13. Initially, the performance characteristics of each asset are determined; sequentially, a number of initial solutions is generated and a portfolio is created, which is optimized. If the results are satisfactory, the simulation is finished; if not, the portfolio is updated.

As a possible drawback, while PSO is considered a straightforward and computationally efficient optimization method, its performance can be affected when used in conjunction with other techniques, such as PT. Consequently, selecting the most appropriate topology and tuning the model parameters - such as the number of particles, inertia coefficient and particle speed - is crucial to obtain optimal results of the PT – PSO combination.

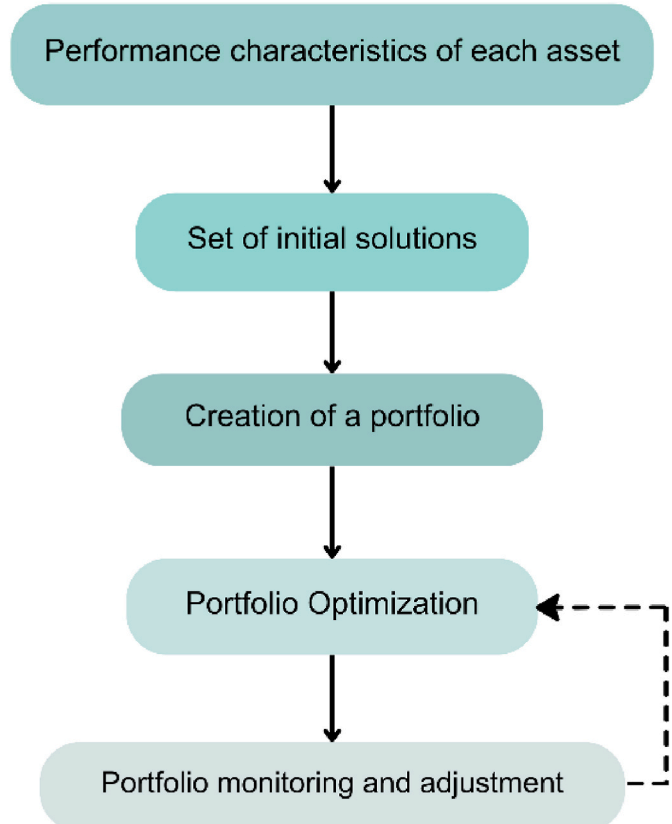

Fig. 13. Proposed methodology.

5. Conclusions

Considering commercial software to design and optimize HRES, HOMER is an important tool, focusing on LCOE and NPC; nevertheless, HOMER has limitations: paid software, need of technical knowledge, influence of the input data quality on the results and a limited range of the components nominal power. PSO is one of the most widely used metaheuristic algorithms applied to the optimization of HRES powered RO systems. Compared to the commercial software, PSO shows more flexibility and has advantages such as accuracy, iteration and program simplicity. Despite these qualities, some gaps are identified in the proposed bibliometric and systematic review, a motivation to improve the quality of PSO results when applied to HRES powered RO plants. One of the most substantial limitations is PSO low convergence when operating with high-dimensional search spaces and the risk of local convergence. Hence, a strategy is to associate other algorithms or software with PSO, to deal with its limitations. As PT is generally applied to create and optimize the best portfolio assets, finding the minimum risk for an expected return, the combination of these two methodologies can be beneficial to improve PSO weakness, improving the results quality. As far as the authors are aware, the combined use of PT and PSO for the energy management of a HRES powered RO plant is innovative.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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